

Parameter	Value	Unit	Parameter	Value	Unit
Model name	NREL5MW	–	Air density	1.225	kg/m ³
Rotor diameter	126	m	Optimal power coefficient $C_{p,\text{opt}}$	0.482	–
Optimal tip speed ratio λ_{opt}	7.55	–	Rotor inertia	4×10^6	kg m ²
Gearbox ratio	97	–	Generator efficiency	0.944	–
Rated power	5×10^6	W	Cut-in wind speed	3.0	m/s
Rated wind speed	11.4	m/s	Cut-out wind speed	25.0	m/s
Rated rotor speed	12.1	rpm	Minimum rotor speed	0.722	rad/s
Pitch rate	10	°/s	Brake torque	28 116.2	N m
Yawing rate	5	°/s	Rated torque	31 465 000	N m
Maximum generator torque	47 402.91	N m			